

**REGULARIZING CONTROL INVARIANCE:
CONTROL BARRIER FUNCTIONS AND WHERE TO FIND THEM**

by

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FOREWORD

At the time of writing this dissertation, the world is going through a significant change driven by artificial intelligence (AI) and autonomous systems. Even as I write these first sentences, I wonder at how long this transformation will persist, and what the world will look like once it's through. In spite of this uncertainty, I remain confident that the relevance of this research field will only increase with time. In a world where the number of answers is rapidly engulfing the number of (meaningful) questions, the veracity of each answer is increasingly difficult to ratify. AI can answer anything but, can it answer correctly? Humans and their civil institutions will ultimately require *certificates* of correctness for the answers to meaningful questions, and certificates of *safety* for any autonomous system on which lives depend. These certificates allow proper routing of responsibility in the event of mistakes or failures; they are the foundation that governments will use to regulate AI and autonomous systems in the years to come.

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ABSTRACT

The control invariance property of sets was characterized in terms of the Bouligand contingent cone by Nagumo in 1942. Under standard regularity and transversality conditions, the expression for the contingent cone simplifies significantly at the boundary. This dissertation demonstrates how control barrier functions extend this characterization to the entire set beyond its boundary: a process termed regularization. In doing so, the dissertation exposes the geometric conditions under which control barrier functions exist. These existence conditions serve as the safety equivalent to the converse Lyapunov theorems for stability. Finally, to bridge theory and application, the dissertation highlights how tools from the reference governor literature can be used to obtain control barrier functions.

The complete dissertation will be available once all associated research is published.